```
sudo su –

curl -fsSL https://get.docker.com -o install-docker.sh

sh install-docker.sh
```

```
df -h
```

Shows **disk space usage** on your system.

- df → Disk filesystem information
- -h → Human readable (MB / GB instead of bytes)

Before running Docker cleanup or creating containers

```
docker  system prune -a
```

Removes **unused Docker data** to free disk space

- docker system prune → Cleans unused Docker resources
- -a → Remove **all unused images**, not just dangling ones

```
docker  system  prune  -a  –volumes
```

removes **everything unused including volumes**.

docker system prune → Cleans unused D ocker resources

-a → Remove **all unused images**, not just dangling ones

--volumes → Also delete unused Docker volumes

```
docker network create selenium-grid
```

Creates a **custom Docker network**.

- Selenium Hub
- Selenium Nodes

To communicate using container names.

```
docker run -d --name selenium-hub --network selenium-grid \
-p 4444:4444 -p 4442:4442 -p 4443:4443 \
```

selenium/hub:4.18.1

Starts Selenium Grid Hub container.

- docker run → Start container
- -d → Run in background (detached mode)
- --name selenium-hub → Container name
- --network selenium-grid → Connect to created network

- 4444 → Web UI + test connection
- 4442 → Event publish port
- 4443 → Event subscribe port

- selenium/hub:4.18.1 → Selenium Grid Hub version

```
docker run -d --name selenium-edge --network selenium-grid \
-e SE_EVENT_BUS_HOST=selenium-hub \
-e SE_EVENT_BUS_PUBLISH_PORT=4442 \
-e SE_EVENT_BUS_SUBSCRIBE_PORT=4443 \
--shm-size="2g" \
selenium/node-edge:4.18.1
```

Starts Edge browser node and connects it to Selenium Hub.

SE_EVENT_BUS_HOST=selenium-hub

Tells node → Hub container name.

SE_EVENT_BUS_PUBLISH_PORT=4442

Port used to send events to hub.

SE_EVENT_BUS_SUBSCRIBE_PORT=4443

Port used to receive events from hub.

4444:4444 – Selenium Grid UI & API

Main **Grid endpoint**

Used by:

- Test scripts
- Browser nodes
- Grid web UI

4442:4442 – Event Bus Publish Port

Nodes **send events** to the hub

Examples:

- Node registration
- Health updates

4443:4443 – Event Bus Subscribe Port

Nodes **receive events** from the hub

Examples:

- Test execution commands

Port	Direction	Purpose
4442	Node → Hub	Publish events
4443	Hub → Node	Subscribe to events

[Test Script]

↓ (4444)

[Selenium Hub]

↓ (4443)

[Browser Node]

↑ (4442)

```
1.df -h
2.docker system prune -a
3.docker system prune -a --volumes
4.docker network create selenium-grid

5.docker run -d --name selenium-hub --network selenium-grid \
-p 4444:4444 -p 4442:4442 -p 4443:4443 \
selenium/hub:4.18.1

6.docker run -d --name selenium-edge --network selenium-grid \
-e SE_EVENT_BUS_HOST=selenium-hub \
-e SE_EVENT_BUS_PUBLISH_PORT=4442 \
-e SE_EVENT_BUS_SUBSCRIBE_PORT=4443 \
--shm-size="2g" \
selenium/node-edge:4.18.1
```